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Key Projects and Experiences

1. *Master's Thesis Project*

Indian Institute of Technology Bombay

1 Aug 2025 - May 2026

(MSc/ Guide: Dr. Dinesh Kabra)

Simulation, Fabrication, and Characterisation of Top Conducting Oxide for Solar Cells.

- Conducting Experiments in the NCPRE (National Centre for Photovoltaic Research and Education) lab.
- optimised sputtered IZO nanolayers for their optoelectronic properties to increase the efficiency of Tandem Solar Cells.
- Used Confocal Sputtering, evaporation for fabrication & UV-Vis-NIR, 4-point probe, Profilometer for Characterisation.
- Simulated multilayer nanostructure IZO/Ag/IZO. Improved optical transparency and device current using admittance matching for EM waves.

2. *Summer Practical Training*

Bhabha Atomic Research Centre

10th May to 20th July 2025

(MSc physics/ Guide: SO/E Dr. Subodha Sahoo)

High-pressure Raman spectroscopic study on the Copper oxide system in HP & SRPD Division, BARC.

- Material $Na_2Cu_2TeO_6$ was investigated under maximum 10 GPa pressure for possible phase transitions.
- Used Diamond Anvil Cell to create high-pressure conditions and analysed using Raman Spectroscopy.

3. *Experimental Work Visit*

Raja Ramanna Centre for Advanced Technology

29th June to 8th July 2025

(MSc physics)

Conducted experiments at the Synchrotron Radiation (INDUS-2) facility, Indore, as BARC team member.

- Worked on Beamline 11 (Extreme Condition XRD Beamline) for a week to collect multiple sample data.
- Observed high-pressure properties of $Na_2Cu_2TeO_6$ with high-intensity X-rays by changing pressure.

4. *Capstone Project*

Lovely Professional University

Nov-2022 to May-2023

(BSc honors / Faculty: Prof. Dr. Mukesh Kumar)

Synthesised PANI(polymer)/ Fe_3O_4 a composite material for the Electromagnetic Interference Shielding.

- Magnetite(Fe_3O_4) was synthesised independently, and then composite nanoparticles were synthesised in situ with polymer PANI(Polyaniline), performing 22 to 27 hours of lab work a week for over 4 months.
- The shielding effectiveness of the prepared material was measured with a test VNA (vector network analyser). Other tests, such as FESEM, XRD, FTIR were performed for structural analysis and imaging.
- Partnered with a professor at GNDU, Amritsar, to analyse Vector Network Analyser (VNA) results.

5. *Anveshna Project*

Lovely Professional University

Jan-Apr 2022

(BSc honors / Faculty: Prof. Dr. Ajit Kumar Sharma)

Microbial fuel Cell- production of hydrogen gas from bio-waste.

- Worked with Ph.D. student Mr. Karan Singh Maan(assistant professor now) to assemble the experiment.
- Collected required chemicals for anaerobic respiration in the fuel cell, presented experiment in Chennai.

Other Projects

1. *Random value generator without microprocessor*

Indian Institute of Technology Bombay

Nov 2024

MSc physics-Electronics course project

- Designed a random signal generator without the use of any microprocessor connected to a sound and light source. To be used in the farms/villages for the protection of rural areas from wild animal interventions.

2. *Scientific Diagrams*

Lovely Professional University

Mar 2023

(BSc honors- self Project)

- Helped design **Scientific Vector Diagrams** for thesis projects of colleagues, using Adobe Illustrator app.

3. *Incredible 8 (Social Service Team lead)*

Lovely Professional University

Jun-August 2022

(BSc honors)

- Led an 8-person team for over two months representing department, directing a comprehensive, multi-phase community support project to provide essential aid and awareness during the coronavirus lockdown.

Conferences and Workshops

- *International Workshop on Semiconductor Epitaxy & Devices for Quantum Technologies* (IIT Bombay 2026)
- *Attended online 'Summer School on Optics & Photonics'* (IIT Delhi 2025)
- *'Fourth International Conference on Recent Advances In Fundamental and Applied Sciences'* (RAFAS 2023)
- Attended online workshop on *Testing Aspects of General Relativity (2023)* arranged by 'IIT Gandhinagar'.

Course Work

1. MSc

August 2024-May 2026

Indian Institute of Technology Bombay

- light matter interaction (quantum optics), Photonics and Engineering, Electromagnetic Theory II, Ultrafast sciences, Programming Lab' 1, Electromagnetic Theory I, Quantum Mechanics I and II, Statistical Physics, Introduction to Condensed Matter Physics, Electronics course and lab.

2. BSc honors

August 2020-April 2023

Lovely Professional University

- Applied Optics, waves and optics, Quantum Mechanics and Applications, Statistical Mechanics, Nuclear and Particle Physics, EM Theory, Modern Physics, Waves and Optics, Analog system, Thermal Physics, Electricity and Magnetism, Mathematical physics I and II, Multivariate Calculus, Algebra and Differential Equations
- Lab and Community Projects:
Waves and Optics laboratory, Solid state physics laboratory, Medical Physics Lab, Electricity and Magnetism lab, Analogue and Digital System labs, Community development project

Technical and Non-Technical Skills

- Programming Language: **Python**- Intermediate level
- Design applications: **Adobe Illustrator, Affinity Designer**- Expert level
- Leading and management qualities .

Achievements/Awards

- *IIT JAM with all India rank 84* (2024), *CUET PG with marks 159/300* (2024).
- 5th position- Inter IIT 7.0 as team member of photography team (2024)
- 3rd prize-Photography competition organised by Pg-cult IIT Bombay (2024)

Positions of Responsibility

- *MSc Representative* - Managing and representing two MSc batches of 120 MSc physics students. (2025-26).
- *Class Representative* - Managing a class of sixty first-year MSc physics students, arranged a farewell involving department funds of **half a lakh rupees** (2024-25).

Others

- Hobbies: Trekking, Sketching, Swimming
- Languages: Marathi, Hindi, English.